THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** |
Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 60°	The angle and its complement add up to 90°.
2	**(b)** 1	Digit sum must be a multiple of 3.
3	**(b)** 80°	Share 180° in the ratio.
4	**(a)** $3\frac{1}{2}$	Convert both to improper fractions first.
5	**(b)** 70°	Find the supplement first, then halve it.
6	**(a)** 4	Count the terms.
7	**(d)** 70°	Exterior angle equals the sum of the two base angles.
8	**(a)** 2	Improper form, then multiply by the reciprocal.
9	**(d)** 60°	They add up to 180°.
10	**(d)** 6174	Largest arrangement minus smallest arrangement.
11	**(d)** 75°	Exterior angle $= \angle A + \angle C$.
12	**(a)** 1	Work from left to right.
13	**(c)** 90°	The angles around a point add up to 360°.
14	**(d)** 9	The first half decides the number.
15	**(a)** 7	Count whole numbers strictly between the difference and the sum.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Let the complement be y; the angle is $y + 20$ and $y + (y + 20) = 90$; $2y = 70$, so $y = 35$ (1); The angle is 55° (1).	1+1
17	Digit sum $= 10 + *$; Must be a multiple of 9, so $* = 8$ (1); Number $= 5814$ (1).	1+1
18	Case 1: 6 is an equal side, base $= 26 - 2 \times 6 = 14$; not valid, since $6 + 6 = 12$ is not greater than 14; Case 2: 6 is the base, equal sides $= (26 - 6) \div 2 = 10$; valid (1); Sides (cm): $10, 10, 6$ (1).	1+1
19	Left to right: $\frac{2}{3} \times \frac{9}{10} = \frac{3}{5}$; Then divide: $\frac{3}{5} \times \frac{5}{3}$ (1); Answer 1 (1).	1+1
20	Larger + smaller $= 180$ and larger − smaller $= 20$; Larger $= (180 + 20) \div 2 = 100$ (1); The angles are 100° and 80° (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Adjacent angles form a linear pair: 5 parts $= 180^\circ$ (1); One part $= 36^\circ$, so the adjacent angles are 36° and 144° (1); Vertically opposite angles are equal, so the four angles are 36°, 144°, 36°, 144° (1).	1+1+1
22	$7621 - 1267 = 6354$; $6543 - 3456 = 3087$; $8730 - 0378 = 8352$; $8532 - 2358 = 6174$ (1); Steps needed $= 4$ (2).	1+2
23	One part $= 180^\circ \div 9 = 20^\circ$; Angles 40°, 60°, 80° (1); Exterior angle $= 180^\circ - 80^\circ$ (1); $= 100^\circ$ (1).	1+1+1
24	Area $= \frac{3}{2} \times \frac{9}{4}$; $= 3\frac{3}{8}$ m^2 (1); Perimeter $= 2(1\frac{1}{2} + 2\frac{1}{4})$; $= 2 \times 3\frac{3}{4}$ (1); $= 7\frac{1}{2}$ m (1).	1+1+1
	OR: Let the complement be y; the angle is $2y + 6$ and $y + (2y + 6) = 90$ (1); $3y = 84$, so $y = 28$ (1); The angle is 62° and its complement is 28° (1).	1+1+1
25	$11(a + b) = 99$, so $a + b = 9$; $a - b = 3$ (1); $a = 6$, $b = 3$ (1); Number $= 63$ (1).	1+1+1
	OR: Equal sides, so $\angle Q = \angle R$; $3x - 10 = x + 6$ so $x = 8$ (1); $\angle Q = \angle R = 14^\circ$; $\angle P = 180^\circ - 28^\circ = 152^\circ$ (1); Exterior angle at P $= 28^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	They are supplementary: 5 parts $= 180^\circ$ (1); One part $= 36^\circ$, so the angles are 72° and 108° (2); Linear pair with the larger: $180^\circ - 108^\circ = 72^\circ$ (2).	1+2+2

	OR: Sequence: $7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1$ (1); Largest $= 52$ (1); Even terms $= 11$ (1); Odd terms $= 6$ (1); Total terms $= 17$ (1).	1+1+1+1+1
27	Equal sides: $2x + 1 = 3x - 4$, so $x = 5$ (1); Each equal side $= 11$ cm (1); Base $= 5 + 3 = 8$ cm (1); Perimeter $= 11 + 11 + 8 = 30$ cm (1); Existence: $11 + 8 = 19 > 11$ and $11 + 11 = 22 > 8$ (1).	1+1+1+1+1
	OR: Day 1 $= \frac{1}{4} \times 240 = 60$ m^2 (1); Remainder $= 180$ m^2 (1); Day 2 $= \frac{2}{3} \times 180 = 120$ m^2 (1); Day 3 $= 180 - 120$ (1); $= 60$ m^2 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 21	Number = part ÷ fraction.
2	**(b)** 75°	Let the angle be y; its supplement is $y + a$.
3	**(b)** 60	Use the middle number times the count when the count is odd.
4	**(c)** 70°	Substitute d.
5	**(b)** $2\frac{1}{2}$	Time = $1 \div$ rate.
6	**(b)** parallel	Check the angle relation.
7	**(c)** even	Test with small values of n.
8	**(b)** SAS	Check which parts are given and where they sit.
9	**(a)** $\frac{3}{4} \div \frac{1}{8}$	Compare by computing.
10	**(d)** 100°	Larger + smaller = 180° and larger − smaller = d.
11	**(a)** 3 but not 9	Use the digit-sum tests.
12	**(c)** 24	Add the three sides.
13	**(b)** 105	Distance = rate × quantity.
14	**(c)** 60°	Share 90° in the given ratio.
15	**(a)** 99	Reverse the digits, then add.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Middle odd number = $51 \div 3 = 17$; Others differ by 2: 15 and 19 (1); Numbers: $15, 17, 19$; Check sum = 51 (1).	1+1
17	Exterior = sum of the interior opposite angles; $4x - 10 = 3x + 30$ so $x = 40$ (1); Exterior angle = 150° (1).	1+1
18	Number = $27 \div \frac{3}{4}$; = 36 (1); $\frac{1}{4}$ of it = 9 (1).	1+1
19	Compare 80° and 80° (1); Equal alternate interior angles, so the lines are parallel (1).	1+1
20	$39 + 93 = 132$; $132 + 231 = 363$ (1); Steps = 2; palindrome 363 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Spent = $\frac{1}{4} \times 240 = 60$; Left = $240 - 60 = 180$ (1); Each = $180 \div 6$ (1); = ₹30 (1).	1+1+1
22	4 parts = 180°, one part = 45° (1); The angles are 45° and 135° (1); Complement of the smaller angle = $90^\circ - 45^\circ = 45^\circ$ (1).	1+1+1
23	Middle number = $75 \div 5 = 15$; First number = $15 - 2 = 13$ (1); Numbers: $13, 14, 15, 16, 17$ (1); Check sum = 75 (1).	1+1+1
24	Corresponding sides of congruent triangles are equal, so $AB = DE$; $5x - 2x = 18 - 3$, so $3x = 15$ and $x = 5$ (1); $DE = AB = 28$ cm; Perimeter = $AB + BC + AC = 28 + 7 + 9$ (1); = 44 cm (1).	1+1+1
	OR: Bracket: $\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$ (1); Multiply: $\frac{5}{6} \times \frac{6}{5}$ (1); = 1 (1).	1+1+1
25	Adjacent angles form a linear pair: 9 parts = 180° (1); One part = 20°, so the adjacent angles are 20° and 160° (1); Vertically opposite angles are equal, so the four angles are 20°, 160°, 20°, 160° (1).	1+1+1
	OR: $9810 - 0189 = 9621$; $9621 - 1269 = 8352$; $8532 - 2358 = 6174$ (1); Steps needed = 3 (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Total parts = 10 and $360^\circ \div 10 = 36^\circ$ (1); The angles are 36°, 72°, 108°, 144° (2); Kinds: 36° acute, 72° acute, 108° obtuse, 144° obtuse (2).	1+2+2
	OR: Palindrome form $1b1$ with digit sum = $2 \times 1 + b$ (1); Need $2 + b$ to be a multiple of 3 (1); Possible b: $1, 4, 7$ (1); Palindromes: $111, 141, 171$ (2).	1+1+1+2

27	$2, 4, 6$: 2+4=6 not > 6, no triangle; $2, 4, 7$: 2+4=6 not > 7, no triangle; $2, 6, 7$: 2+6=8 > 7, triangle; $4, 6, 7$: 4+6=10 > 7, triangle (2); Number of triangles $= 2$ (3).	2+3
	OR: Syrup $= \frac{2}{5} \times \frac{5}{2} = 1$ L (1); Water $= 2\frac{1}{2} - 1 = 1\frac{1}{2}$ L (1); Glasses $= 2\frac{1}{2} \div \frac{1}{4}$ (1); $= 10$ (1); Full glasses: 10 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 110°	The other acute angle is $90^\circ - a$; the exterior angle is supplementary to it.
2	**(d)** 3	Convert both to improper fractions first.
3	**(c)** 75°	The angle and its complement add up to 90°.
4	**(b)** 2	Digit sum must be a multiple of 3.
5	**(d)** 90°	Share 180° in the ratio.
6	**(c)** 6	Improper form, then multiply by the reciprocal.
7	**(b)** 60°	Find the supplement first, then halve it.
8	**(d)** 15	Count the terms.
9	**(a)** 100°	Exterior angle equals the sum of the two base angles.
10	**(a)** $\frac{2}{3}$	Work from left to right.
11	**(d)** 72°	They add up to 180°.
12	**(b)** 9261	Largest arrangement minus smallest arrangement.
13	**(a)** 65°	Exterior angle $= \angle A + \angle C$.
14	**(b)** 30	Number $=$ part $\div$ fraction.
15	**(a)** 85°	The angles around a point add up to 360°.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$\angle B = 180^\circ - 110^\circ = 70^\circ$; $\angle C = \angle B = 70^\circ$ (equal sides) (1); $\angle A = 180^\circ - 2(70^\circ) = 40^\circ$ (1).	1+1
17	Product $= 1\frac{1}{2}$; Compare with 2\frac{1}{2} (1); The product is smaller; Because the multiplier is less than 1 (1).	1+1
18	Let the complement be y; the angle is $y + 40$ and $y + (y + 40) = 90$; $2y = 50$, so $y = 25$ (1); The angle is 65° (1).	1+1
19	Digit sum $= 15 + *$; Must be a multiple of 9, so $* = 3$ (1); Number $= 6237$ (1).	1+1
20	Case 1: 10 is an equal side, base $= 22 - 2 \times 10 = 2$; valid; Case 2: 10 is the base, equal sides $= (22 - 10) \div 2 = 6$; valid (1); Sides (cm): $10, 10, 2$ or $6, 6, 10$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	One part $= 180^\circ \div 6 = 30^\circ$; Angles 30°, 60°, 90° (1); Exterior angle $= 180^\circ - 30^\circ$ (1); $= 150^\circ$ (1).	1+1+1
22	Area $= \frac{7}{3} \times \frac{3}{2}$; $= 3\frac{1}{2}$ m^2 (1); Perimeter $= 2(2\frac{1}{3} + 1\frac{1}{2})$; $= 2 \times 3\frac{5}{6}$ (1); $= 7\frac{2}{3}$ m (1).	1+1+1
23	Let the complement be y; the angle is $2y + 15$ and $y + (2y + 15) = 90$ (1); $3y = 75$, so $y = 25$ (1); The angle is 65° and its complement is 25° (1).	1+1+1
24	$11(a + b) = 121$, so $a + b = 11$; $a - b = 1$ (1); $a = 6$, $b = 5$ (1); Number $= 65$ (1).	1+1+1
	OR: Equal sides, so $\angle Q = \angle R$; $3x - 20 = x + 10$ so $x = 15$ (1); $\angle Q = \angle R = 25^\circ$; $\angle P = 180^\circ - 50^\circ = 130^\circ$ (1); Exterior angle at P $= 50^\circ$ (1).	1+1+1
25	Spent $= \frac{2}{5} \times 360 = 144$; Left $= 360 - 144 = 216$ (1); Each $= 216 \div 9$ (1); $=$ ₹24 (1).	1+1+1
	OR: 9 parts $= 180^\circ$, one part $= 20^\circ$ (1); The angles are 40° and 140° (1); Complement of the smaller angle $= 90^\circ - 40^\circ = 50^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	They are supplementary: 6 parts $= 180^\circ$ (1); One part $= 30^\circ$, so the angles are 30° and 150° (2); Linear pair with the larger: $180^\circ - 150^\circ = 30^\circ$ (2).	1+2+2

	OR: Sequence: $20, 10, 5, 16, 8, 4, 2, 1$ (1); Largest $= 20$ (1); Even terms $= 6$ (1); Odd terms $= 2$ (1); Total terms $= 8$ (1).	1+1+1+1+1
27	Equal sides: $2x + 2 = 3x - 3$, so $x = 5$ (1); Each equal side $= 12$ cm (1); Base $= 5 + 6 = 11$ cm (1); Perimeter $= 12 + 12 + 11 = 35$ cm (1); Existence: $12 + 11 = 23 > 12$ and $12 + 12 = 24 > 11$ (1).	1+1+1+1+1
	OR: Day 1 $= \frac{2}{5} \times 180 = 72$ m^2 (1); Remainder $= 108$ m^2 (1); Day 2 $= \frac{1}{3} \times 108 = 36$ m^2 (1); Day 3 $= 108 - 36$ (1); $=$ 72 m^2 (1).	1+1+1+1+1

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